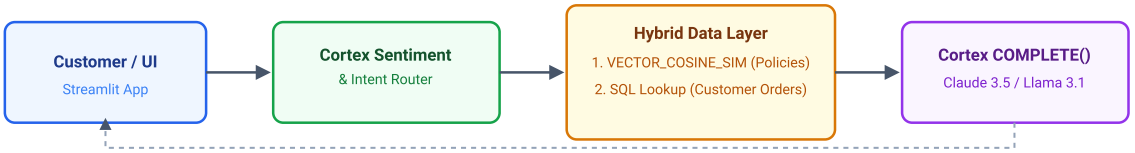
E-Grocery Support Agent Architecture

Functional Specification & Technical Flow Diagram — Streamlit in Snowflake (SiS) & Cortex AI

SYSTEM ARCHITECTURE	PRIMARY AI ENGINE	VECTOR EMBEDDING	SECURITY PROTOCOL
Native Cortex RAG	**Snowflake Cortex LLM**	**e5-base-v2 (768-d)**	**Snowflake Native RBAC**

1. HIGH-LEVEL END-TO-END SYSTEM ARCHITECTURE

The solution operates entirely within the security boundary of Snowflake. User queries pass through Streamlit in Snowflake (SiS), triggering transactional DB queries and vector search over Cortex AI embedded policy chunks.



2. KEY SYSTEM FUNCTIONAL PILLAR MODULES

A. Automated RAG & Parsing
Unstructured PDFs/Markdown files stored in stages are parsed via `PARSE_DOCUMENT()` and vector-indexed with `e5-base-v2` embeddings for accurate grounding.

B. Transactional Sync
Direct SQL integration joins live customer order states (`CUSTOMER_ORDERS`) with vector policy context, enabling instantaneous item eligibility verification.

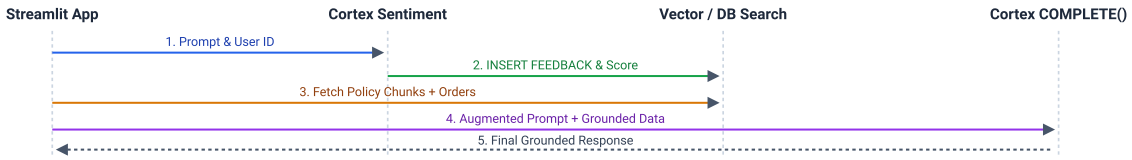
C. Real-time Feedback Loop
Every message undergoes sentiment analysis (`SENTIMENT()`). Scores and categories are recorded directly into `CUSTOMER_FEEDBACK` tables for auditing.

3. EXECUTION FLOW OVERVIEW

When a user sends an inquiry, the system executes a four-phase synchronous routine: (1) Sentiment & Intent analysis, (2) Parallel extraction of user order records and top vector-matched policy chunks, (3) Prompt assembly with strict grounding rules, and (4) Context-aware response generation via Cortex LLM.

4. DETAILED COMPONENT & PIPELINE SEQUENCE FLOW

The sequence diagram below models the synchronous event flow between the user interface, Snowflake Cortex functions, and internal warehouse tables.



5. SNOWFLAKE CORE DATA SCHEMA ARCHITECTURE

Table Name	Type / Usage	Key Columns	Functional Description
`POLICY_DOCUMENT_CHUNKS`	Vector Store	`DOC_NAME`, `CHUNK_TEXT`, `CHUNK_EMBEDDING`	Stores text chunks and 768-dimensional vector embeddings generated by Cortex `EMBED_TEXT_768`.
`CUSTOMER_ORDERS`	Transactional DB	`ORDER_ID`, `CUSTOMER_ID`, `STATUS`, `ORDER_DATE`	Contains real-time purchase logs used to ground return eligibility and delivery inquiries.
`CUSTOMER_FEEDBACK`	Analytics Log	`MESSAGE`, `CATEGORY`, `SENTIMENT_SCORE`	Captures audit history and Cortex sentiment scores (-1.0 to +1.0) for every user interaction.

6. LOGIC MATRIX: INTENT CATEGORIZATION & EXECUTION

Policy & General Inquiry
Trigger: Shipping rules, perishable policies, cancellations.
Action: Vector similarity search against `POLICY_DOCUMENT_CHUNKS`. Returns citation sources with answer.

Return & Refund Workflow
Trigger: Missing item, spoiled produce, refund request.
Action: Checks order timestamp against 48-hr perishable return policy window. Calculates refund amount.

7. SECURITY, GOVERNANCE & HUMAN ESCALATION GUARDRAILS

To operate safely within enterprise parameters, the chatbot implements strict data isolation and quality control mechanisms directly at the Snowflake engine layer.

Security & RBAC Enforcement

- **Role-Based Access:** Streamlit session inherits active Snowflake user roles, preventing access to unauthorized customer accounts.
- **Zero Data Leakage:** All vector operations and LLM inferences occur strictly inside Snowflake's boundary. No data is sent to external APIs.
- **SQL Injection Shielding:** Queries sanitize inputs using string escaping parameters.

Human Escalation Trigger Rules

- **Negative Sentiment Threshold:** If Cortex `SENTIMENT()` score falls below `-0.5`, the prompt injects an immediate human agent transfer offer.
- **Uncertainty Fallback:** If vector search cosine similarity score is below `0.35`, the system flags lack of context rather than guessing.
- **Order Dispute Limit:** Refunds exceeding $50 automatically trigger manual approval flags.

8. KEY OPERATIONAL METRICS & SUCCESS TRACKING

The system actively measures operational efficiency across deflection, speed, and precision metrics:

Metric Category	Target KPI	Validation Mechanism
Ticket Deflection Rate	> 40% Reduction	Percentage of tier-1 inquiries resolved without human representative intervention.
Response Latency	< 3.0 Seconds	Measured end-to-end execution time from query input to Streamlit response render.
Hallucination Rate	0% (Strict Grounding)	System prompt explicitly forbids outside knowledge; forced citation requirements.
Customer Sentiment Impact	+0.25 Avg Improvement	Monitored via `CUSTOMER_FEEDBACK` sentiment trends over time.

9. DEPLOYMENT CHECKLIST

- Verify `CORTEX_ENABLED_CROSS_REGION = 'ANY_REGION'` parameter is active in target account.
- Upload initial PDF/Markdown policy files into `@ONLINE_POLICY_STAGE` and execute chunking script.
- Grant `SELECT` and `INSERT` privileges on support schema tables to Streamlit app runtime role.